# Origin of Replication (ORI) & Cloning

- ORI is the specific sequence in a DNA which is responsible for initiation of replication.
- Thus, an alien DNA is linked with the ORI, so that it can replicate in the host organism. This is called as cloning.
- ✳ Cohen & Boyer for the first time in 1972 established this by isolating the antibiotic resistance gene by cutting the DNA piece from an antibiotic-resistant plasmid using restriction enzymes.

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- This plasmid DNA acts as vector DNA to transfer the piece of DNA attached to it.
- Antibiotic resistance gene was linked with plasmid vector using DNA ligase.
- This DNA is called recombinant DNA.
- When this DNA was transferred in E. coli it could replicate using the new host's DNA polymerase and create multiple copies. This is known as cloning.

# Restriction Enzymes

- Also called as molecular enzymes, they help in cutting the DNA at specific sites.
- They cut at specific positions called as recognition sites.

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|---|---|
| i) Exonucleases are those enzymes that cut from the end. | ii) Endonucleases are those enzymes that cut at a specific palindromic sequence. |

- They form recognition sites called sticky ends, which are complementary counterparts and can form H-bondings.
- The DNA fragments have the same type of sticky ends, & can be joined together by DNA ligase.

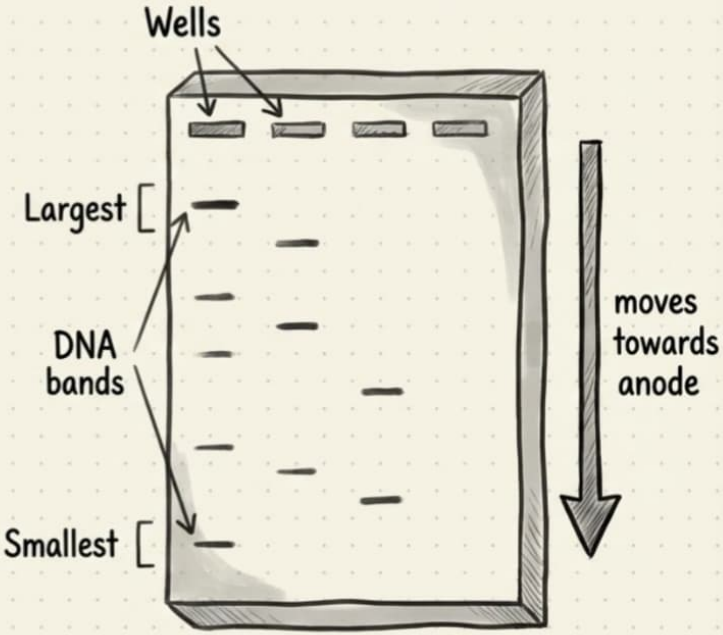
# Gel Electrophoresis

- Cutting of DNA produces DNA fragments which can be separated by gel electrophoresis.
- They are separated under a charged field where DNA fragments being negatively charged (-vely) move towards the anode in an agarose medium.
- They separate according to their sizes through a sieving effect produced by the agarose gel. Thus, the smaller the fragment size, the farther it will move.
- DNA fragments are then stained by Ethidium bromide & followed by UV exposure.
- The separated bands then get cut out from the gel & extracted by this process called elution.

resolve

UV radiation

cloning vectors



## Insertional Inactivation

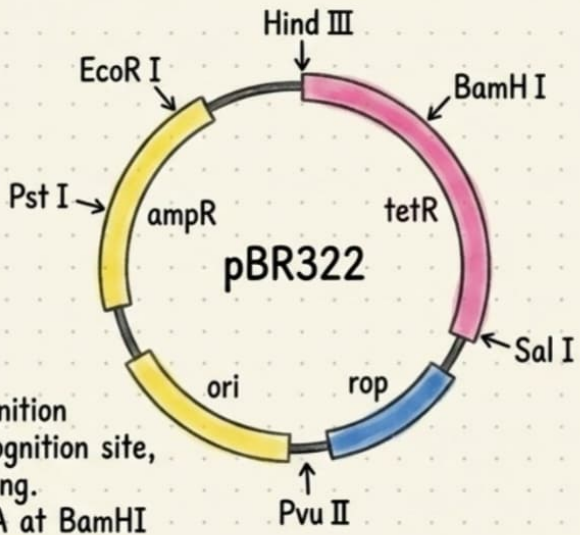
- In insertional inactivation we add desired genes at the resistant selectable markers, then the plasmid will no longer stay resistant for that particular antibiotic.
- e.g., A DNA was inserted in beta-gal, which earlier was producing blue colonies, was now producing white colonies, due to insertion of a recombinant.

## Vectors :-

- Agrobacterium tumefaciens was able to deliver T-DNA to transform plant cells into tumors and these tumors produce chemicals required by pathogens.
- Retro viruses can also convert normal cells to cancerous cells.
- However, later the Ti of Agrobacterium tumefaciens has been modified to a cloning vector that is no more pathogenic but can still be used to transfer GOI (Gene of Interest).
- Similarly, retro viruses have also been disarmed & used as cloning vectors.

# Cloning Vectors

*** ORI :-
- Replication starts from here.
- Also responsible for copy no. of DNA.

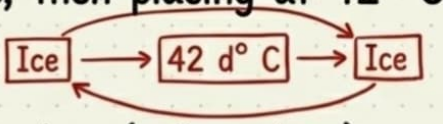



*** Cloning sites :-
- Vector should have only 1 recognition site, as due to more than 1 recognition site, it will complicate the gene cloning.
- e.g., If we ligate a foreign DNA at BamHI site of tetR gene in the vector pBR322, in this way the recombinant plasmid will lose tetR due to insertion of foreign DNA & can be selected from non-recombinants.

*** Selectable markers :-
- Helps in identifying & eliminating non-transformants & selectively permitting the growth of the transformants.
- Transformation is a procedure through which a piece of DNA is introduced in a host bacterium. e.g., ampicillin, chloramphenicol, tetracycline, etc. are markers for E. coli.
- However, normal E. coli do not carry resistance.

## Competent Host

- Host is made competent to take up foreign DNA by treating it with $Ca^{2+}$ ions.
- Recombinant DNA is then forced into such cells by ice, then placing at 42° C (heat shock), & then putting back into ice.




- Other ways are micro-injection, in which DNA is injected in (cell nucleus).
- Gene gun - cells are bombarded with high velocity micro-particles of gold or tungsten coated with DNA.
- (Disarmed pathogens also act to allow to infect cell, transfer recombinant DNA to host).

## * Isolation of DNA :-

- Bacterial / plant / animal cells are treated with enzymes like lysozyme, cellulase, chitinase.
- RNA is removed by ribonuclease & protein by Protease.
- Other molecules can be removed by addition of chilled ethanol.

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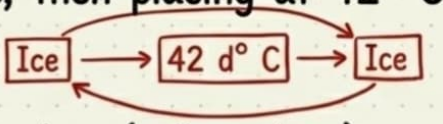


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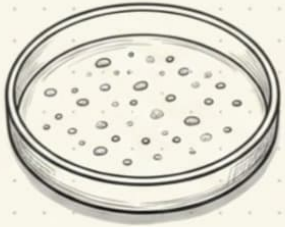
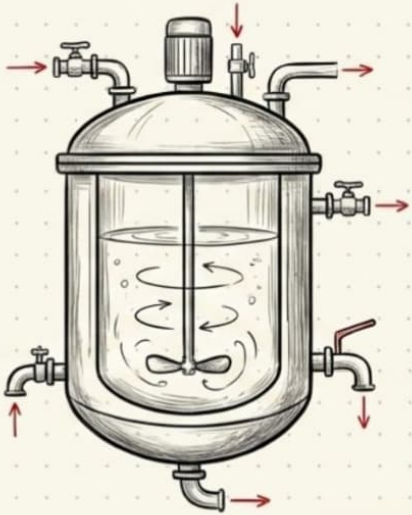
## *** Insertion of recomb DNA into host :-

- If we spread the transformed cells on agar plates, only they will grow and normal ones will die.



## *** Obtaining Gene product :-

- If any protein encoding gene is expressed in host, it is called recombinant proteins.
- They are then cultured & purified.

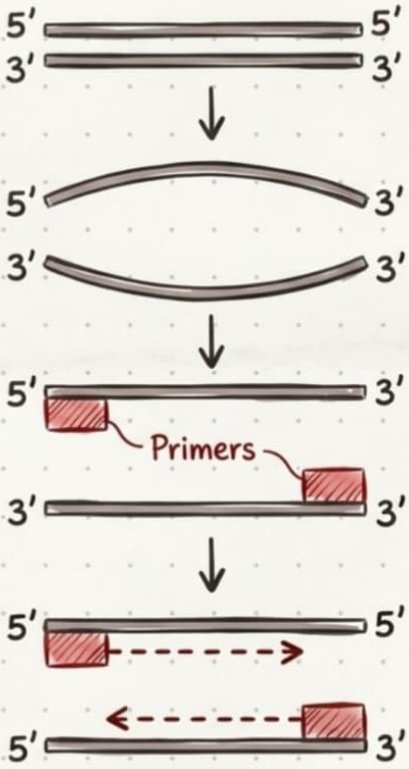


## *** Bioreactors :-

- To produce large quantity of products, Bioreactors are used where large volumes of genes can be cultured.

## * Cutting of DNA :-

- Vector DNA undergoes gel electrophoresis.
- Vector DNA & GOI is mixed with ligase.

## * PCR :-

- Multiple copies of the Gene of interest are synthesised using 2 sets of primers & DNA polymerase.
- Enzyme extends these primers using the nucleotides & DNA as template.
- The repetition is carried on by thermostable DNA, which remains active when high temp is induced during denaturation.
- Amplified strands can now be used to ligate with a vector for cloning.